

PRACTICAL NO 1

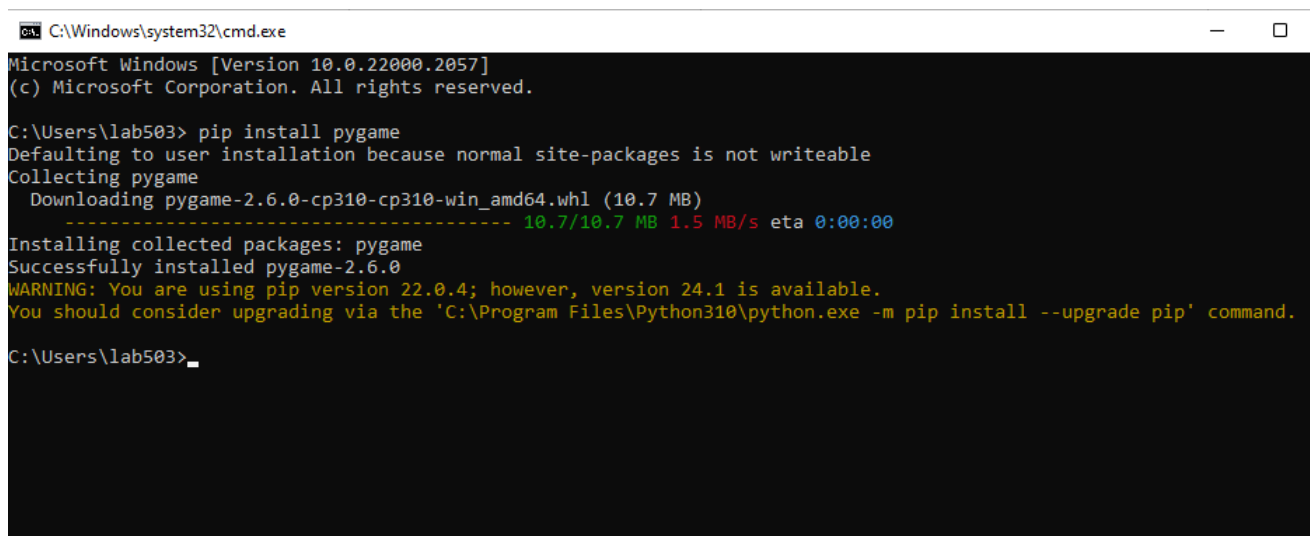
Aim : Learn Basic Game Design Techniques with pygame.

- Create Window / Display
- Circle
- Rectangle
- Square
- Trinagle

A Create Window / Display

Step1: Install pygame

Open command prompt  pip install pygame



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22000.2057]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lab503> pip install pygame
Defaulting to user installation because normal site-packages is not writeable
Collecting pygame
  Downloading pygame-2.6.0-cp310-cp310-win_amd64.whl (10.7 MB)
----- 10.7/10.7 MB 1.5 MB/s eta 0:00:00
Installing collected packages: pygame
Successfully installed pygame-2.6.0
WARNING: You are using pip version 22.0.4; however, version 24.1 is available.
You should consider upgrading via the 'C:\Program Files\Python310\python.exe -m pip install --upgrade pip' command.

C:\Users\lab503>
```

CODE:

```
import pygame, sys
from pygame.locals import *
```

```
pygame.init()
```

```
WINDOW_WIDTH = 1000
```

```
WINDOW_HEIGHT = 500
```

```
WINDOW = pygame.display.set_mode((WINDOW_WIDTH, WINDOW_HEIGHT))
```

```
# Keep window open until closed
```

```
running = True
```

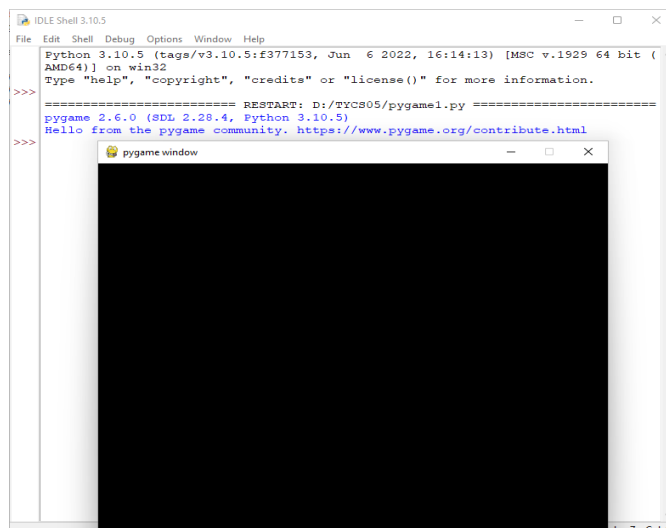
```
while running:
```

```
    for event in pygame.event.get():
```

```
        if event.type == QUIT:
```

```
            running = False
```

```
pygame.quit()
```



B. CODE FOR CIRCLE :

```
import pygame
from pygame.locals import *

pygame.init()

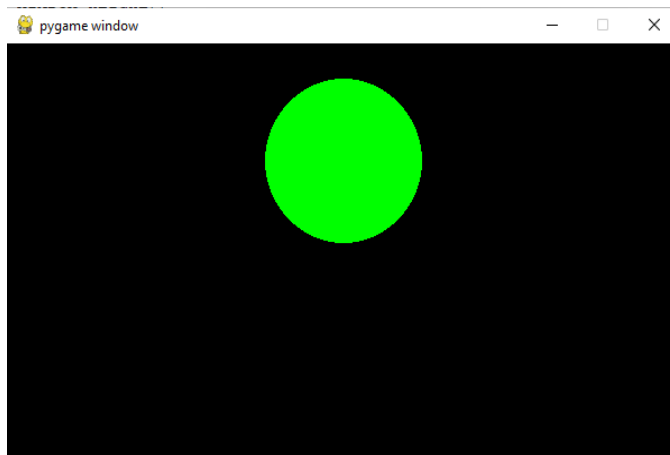
window = pygame.display.set_mode((600, 500))
pygame.display.set_caption("Circle")

pygame.draw.circle(window, (147, 35, 165), (300,250), 150)
pygame.display.flip()

# Keep window open until closed
running = True
while running:
    for event in pygame.event.get():
        if event.type == QUIT:
            running = False

pygame.quit()
```

OUTPUT:



C. CODE FOR RECTANGLE

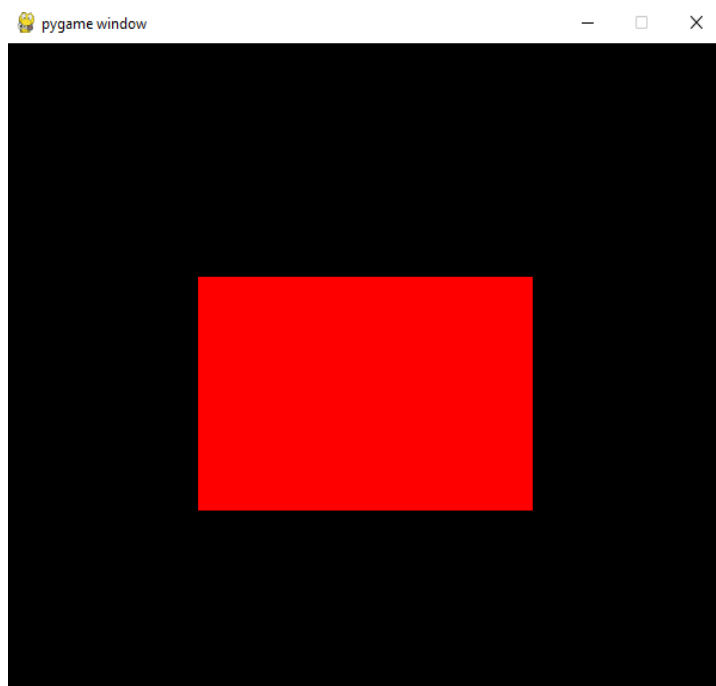
```
import pygame, sys
from pygame.locals import *

pygame.init()

WINDOW_WIDTH = 600
WINDOW_HEIGHT = 500
WINDOW = pygame.display.set_mode((WINDOW_WIDTH,WINDOW_HEIGHT))

pygame.draw.rect(WINDOW,(255,0,0),(160,180,280,180))
pygame.display.flip()
# Keep window open until closed
running = True
while running:
    for event in pygame.event.get():
        if event.type == QUIT:
            running = False
pygame.quit()
```

OUTPUT:



D. CODE FOR SQUARE

```
import pygame
from pygame.locals import *

pygame.init()

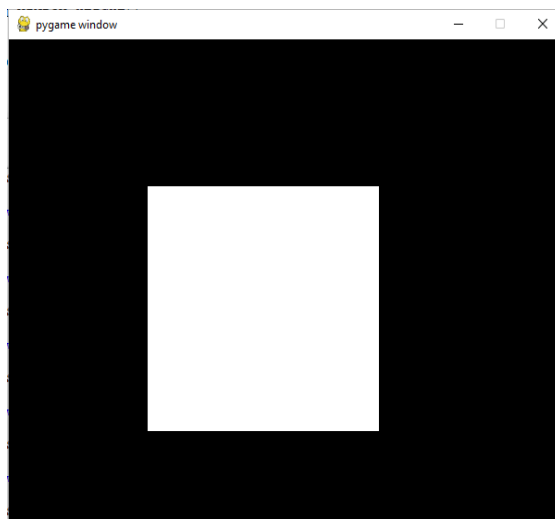
window = pygame.display.set_mode((600, 500))
pygame.display.set_caption("Square")

pygame.draw.rect(window, (255, 255, 255), (150, 150, 250, 250)) # White square
pygame.display.flip()

running = True
while running:
    for event in pygame.event.get():
        if event.type == QUIT:
            running = False

pygame.quit()
```

OUTPUT :



E. Code for Triangle

CODE:

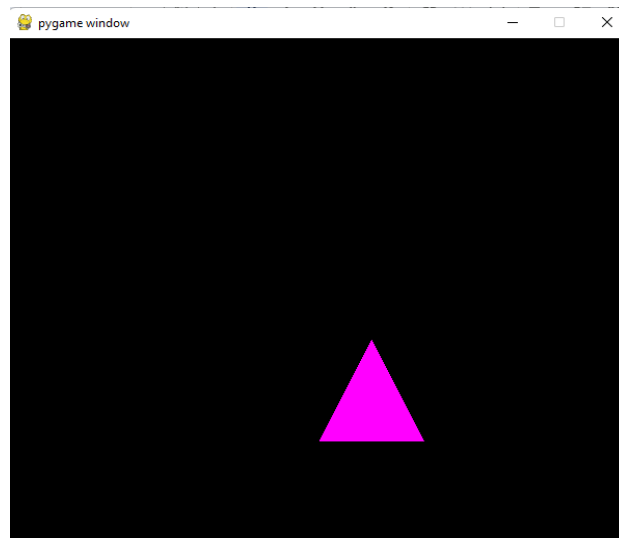
```
import pygame, sys
from pygame.locals import *
pygame.init()
WINDOW_WIDTH = 600
WINDOW_HEIGHT = 500
WINDOW = pygame.display.set_mode((WINDOW_WIDTH, WINDOW_HEIGHT))
pygame.display.set_caption("Triangle")

pygame.draw.polygon(WINDOW, (255, 0, 255), [[400, 400], [350, 300], [300, 400]])
pygame.display.flip()

running = True
while running:
    for event in pygame.event.get():
        if event.type == QUIT:
            running = False

pygame.quit()
```

OUTPUUT



F. CODE FOR SIMPLE HOUSE

CODE:

```
import pygame

# Initialize Pygame
pygame.init()

# Screen dimensions
SCREEN_WIDTH = 800
SCREEN_HEIGHT = 600
screen = pygame.display.set_mode((SCREEN_WIDTH, SCREEN_HEIGHT))
pygame.display.set_caption("Simple House")

# Colors
```

```
WHITE = (255, 255, 255)
BLACK = (0,0,0)
RANDOM = (123, 146, 176)
RED = (45,54,65)

# Fill the background
screen.fill(WHITE)

# Draw the house body (rectangle)
# (x, y, width, height)
pygame.draw.rect(screen, RANDOM, (300, 300, 200, 200))
pygame.draw.rect(screen, BLACK, (350, 400, 80, 100))

# Draw the roof (triangle)
# (surface, color, point1, point2, point3)
pygame.draw.polygon(screen, RED, [(300, 300), (400, 200), (500, 300)])

# Update the display
pygame.display.flip()
# Game loop
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

# Quit Pygame
pygame.quit()
```

OUTPUUT

